FRAUD DETECTION DATA ANALYSIS SYSTEM

Financial Transaction Risk Assessment

Daniyal Nadeem FA24-BBD-120

Dataset: PaySim Synthetic Financial Dataset

Source: Kaggle | kaggle.com/datasets/ealaxi/paysim1

Big Data Analytics — Project Report

6,362,620	8,213	44,945
Total Transactions	Fraudulent Transactions	Outliers Detected

1. Executive Summary

In this report, we have detected fraud in PaySim synthetic financial transaction data in detail. Python (Pandas and NumPY) was used to analyze the data in six organized stages, including data exploration, data cleaning, outlier analysis, grouping users, scoring fraud, and summary statistics with correlation analysis.

PaySim data is a simulated 30 days mobile money transaction dataset that includes maliciously introduced fraudulent behavior, and hence is a standard dataset in financial fraud detection studies. It analyzed 8,213 frauds representing only 0.13% of all the records - imbalanced distribution of classes - and constructed a composite risk score of a fraud to represent multi-signal transaction risk.

Key Findings at a Glance

Finding	Value
Total Transactions Analyzed	**6,362,620**
Fraudulent Transactions	**8,213 (0.13%)**
Outliers Detected (Z-Score > 3)	**44,945 (0.71%)**
Outliers That Are Fraudulent	**1,686 (20.53% of outliers)**
High-Risk Users Identified	**1,542,928**
Balance Mismatches (Sender)	**5,000,784**
Accounts Completely Drained	**1,520,581**
Destination Balance Not Updated	**2,439,427**

2. Dataset Overview

2.1 Source and Description

The PaySim data is a work of Edgar Lopez-Rojas (MLG, Blekinge Institute of technology) and it is publicly accessible on Kaggle. It models transactions that occur in mobile money by a multinational mobile financial service provider within 30 days (744 steps/hour). Bad actors are deliberately inserted in order to model the fraudulent behavior in the real world.

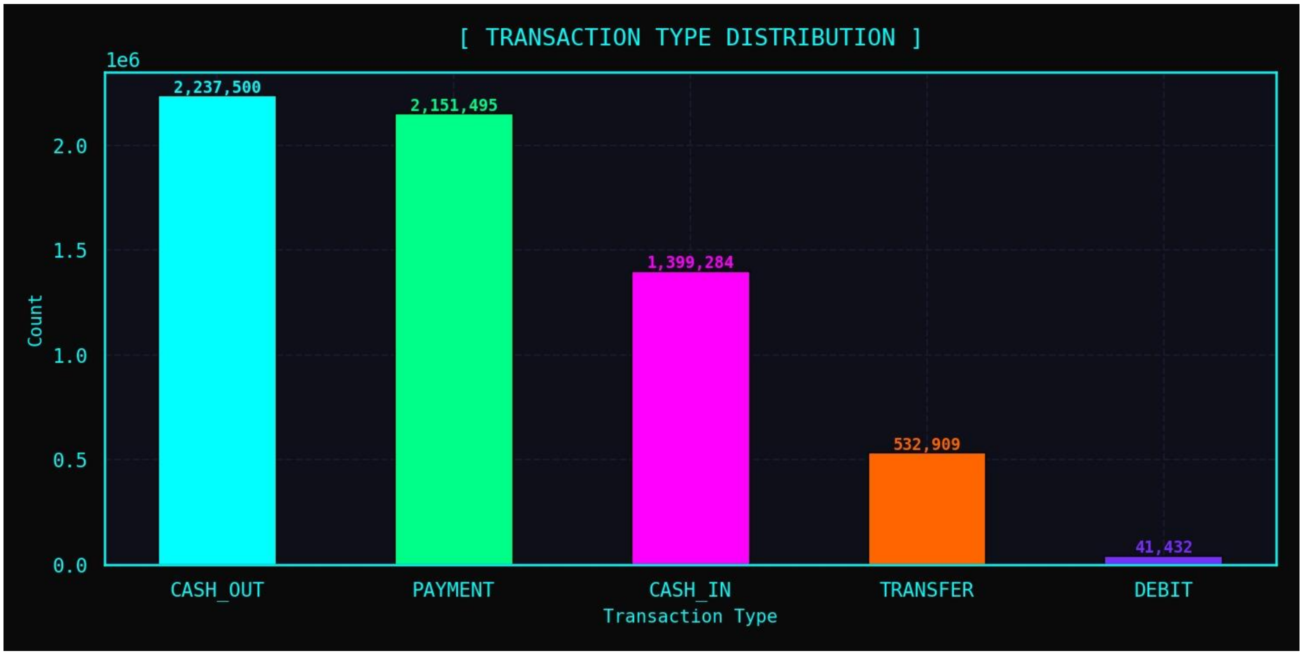
Attribute	Detail
File Name	PS_20174392719_1491204439457_log.csv
Total Records	6,362,620 rows
Total Columns	11
Simulation Period	30 days (744 steps, 1 step = 1 hour)
Creator	Edgar Lopez-Rojas, Blekinge Institute of Technology

2.2 Column Reference

Column	Type	Description
step	Integer	Unit of time; 1 step = 1 hour; max 744
type	String	Transaction type: CASH-IN, CASH-OUT, DEBIT, PAYMENT, TRANSFER
amount	Float	Transaction amount in local currency
nameOrig	String	Unique identifier of the sender
oldbalanceOrg	Float	Sender balance before the transaction
newbalanceOrig	Float	Sender balance after the transaction
nameDest	String	Unique identifier of the recipient
oldbalanceDest	Float	Recipient balance before the transaction
newbalanceDest	Float	Recipient balance after the transaction
isFraud	Integer	Ground truth label: 1 = Fraudulent, 0 = Legitimate
isFlaggedFraud	Integer	System flag: 1 = Flagged suspicious by built-in system

2.3 Transaction Type Distribution

Transaction Type	Count	Fraud Relevance
CASH_OUT	2,237,500	Confirmed — fraud occurs in this type
PAYMENT	2,151,495	No fraud detected
CASH_IN	1,399,284	No fraud detected
TRANSFER	532,909	Confirmed — fraud occurs in this type
DEBIT	41,432	No fraud detected



3. Analysis Phases

Phase 1 — Load & Explore Dataset

The PaySim CSV file was loaded into a Pandas DataFrame using pd.read_csv(). Initial structural exploration was performed to understand shape, data types, missing values, duplicates, and fraud distribution.

Findings

Check	Result	Interpretation
Total Rows	6,362,620	Large-scale dataset suitable for statistical analysis
Total Columns	11	All expected columns present
Missing Values	**0**	Dataset is complete — no imputation required
Duplicate Rows	**0**	No duplicate records found
Fraud Count	**8,213**	Fraudulent transactions are a very small minority
Fraud Percentage	**0.13%**	Severely imbalanced dataset
Max Amount	92,445,516	Extreme outliers exist — requires outlier detection

Key Observations

- There are no missing values or duplicates in the dataset, i.e. no imputation was needed.
- The fraud category is very unbalanced with 0.13% where conventional measures of accuracy would not be accurate.
- The 92.4 million maximum transaction is immensely large compared to the mean of 179,861, which shows that there are a number of outliers.
- PAYMENT, CASH-IN, and DEBIT types of transaction are all fraud free.

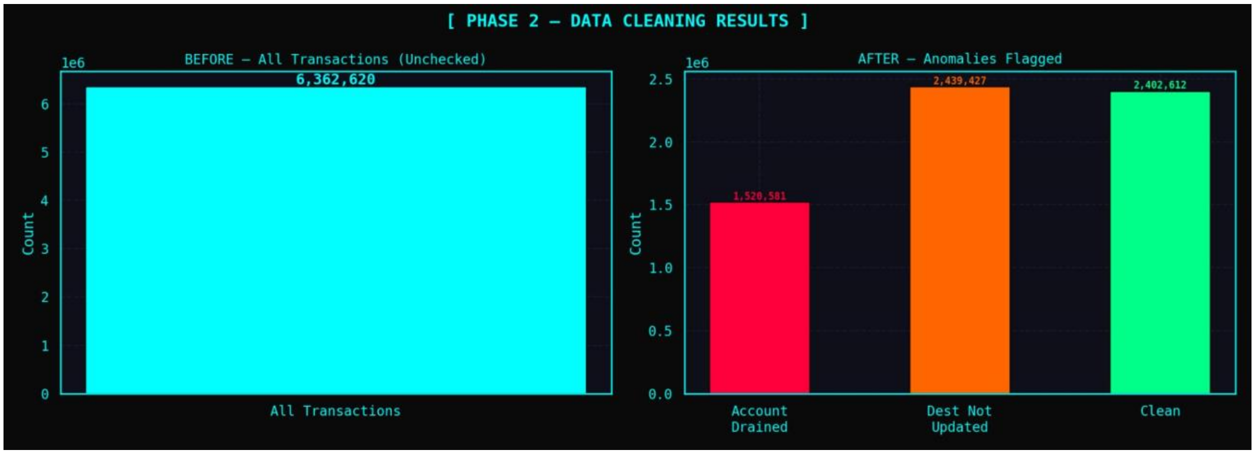
Phase 2 — Data Cleaning & Handling Incorrect Data

Each transaction was checked using balance arithmetic. These transactions in which (oldbalanceOrg - amount) was not equal to newbalanceOrig were labeled as balance mismatches. There was also identification of accounts that became empty, and those destination accounts whose balance had not been updated.

Data Quality Issue	Count	% of Total
Sender Balance Mismatches	**5,000,784**	**78.6%**
Receiver Balance Mismatches	**3,823,834**	**60.1%**
Accounts Completely Drained	1,520,581	23.9%
Destination Balance Not Updated	2,439,427	38.3%

Fraud by Transaction Type

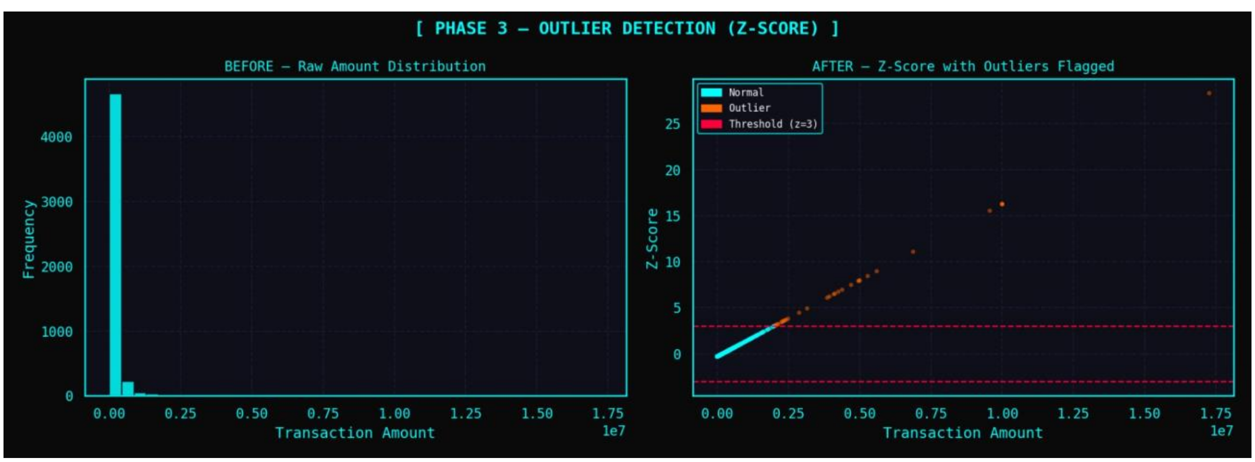
Type	Fraud Count	Note
CASH_OUT	**4,116**	Primary fraud vector
TRANSFER	**4,097**	Second major fraud vector
CASH_IN	0	No fraud detected
PAYMENT	0	No fraud detected
DEBIT	0	No fraud detected



Phase 3 — Outlier Detection Using NumPy Z-Score

NumPy z-scores were used in statistical outlier detection of the column of transaction amount. The statistical unusual transactions were identified based on a z-score of 3 (three standard deviations of the mean).

Z-Score Metric	Value
Mean Transaction Amount	179,861.90
Standard Deviation	603,858.18
Outlier Threshold (Z > 3)	> 1,991,460.31
Total Outliers Detected	44,945 (0.71% of all transactions)
Outliers That Are Fraudulent	**1,686 (20.53% of all outliers)**
Mean Amount of Outliers	4,805,907.67
Max Amount in Dataset	92,445,516.00



Outliers by Transaction Type

Transaction Type	Outlier Count	% of Outliers
TRANSFER	44,084	98.1%
CASH_OUT	861	1.9%

TRANSFER transactions dominate the outlier pool, indicating that large-value wire-like transfers are both the most statistically anomalous and the most fraudulent transaction type in the dataset.

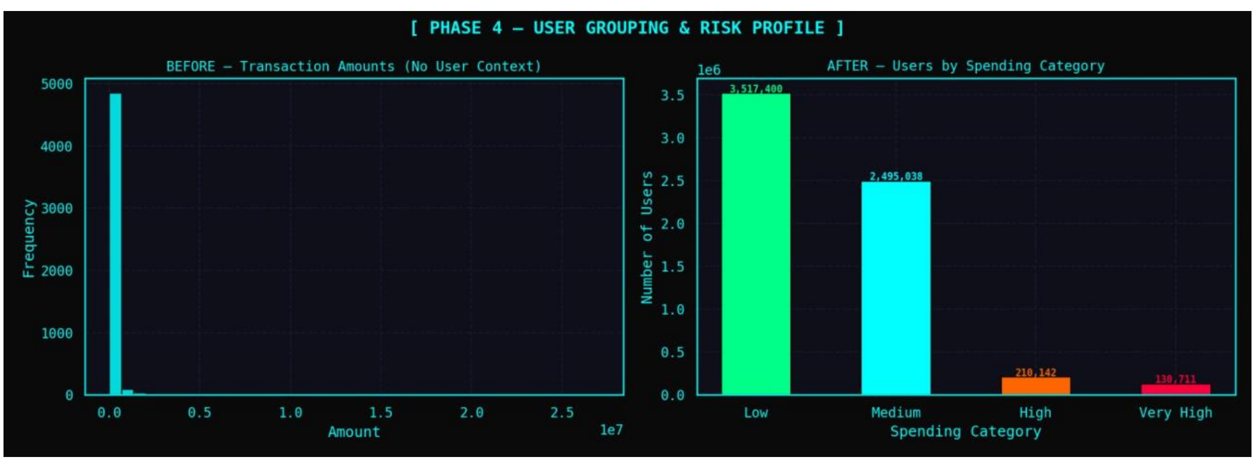
Phase 4 — User Grouping & Spending Behavior Analysis

Transactions were grouped by sender account (nameOrig) to build per-user behavioral profiles. This phase identified high-risk users based on aggregate transaction patterns.

User Metric	Value
Total Unique Users	6,353,307
High-Risk Users Identified	1,542,928

User Behavioral Features Computed

- Total transactions per user
- Total amount spent per user
- Average transaction value
- Maximum and minimum transaction values
- Total fraud transactions attributed to each user
- Total account drain events per user
- Total outlier transactions per user



Phase 5 — Fraud Risk Score (Feature Engineering)

A composite fraud risk score (0–100) was engineered for every transaction by combining five independent risk signals. This approach allows a nuanced, multi-factor assessment rather than relying on any single indicator.

#	Risk Signal	Points	Transactions Affected
1	Risky Transaction Type (TRANSFER or CASH_OUT)	25 pts	2,770,409
2	Outlier Amount (Z-Score > 3)	25 pts	44,945
3	Account Completely Drained	20 pts	1,520,581
4	Destination Balance Not Updated	20 pts	2,439,427
5	System-Flagged as Suspicious	10 pts	16

Risk Score Distribution

Statistic	Risk Score	Interpretation
Mean Score	23.51	Typical transaction is low-risk
Standard Deviation	14.73	Moderate spread
25th Percentile	20.00	Low risk
50th Percentile (Median)	20.00	Majority are single-signal risk
75th Percentile	25.00	Slightly elevated risk
Maximum Score	100	All five signals present

Risk Level Classification

Each transaction was further categorized into one of four risk tiers using pd.cut() on the fraud_risk_score column. This classification enables prioritized human review of the highest-risk transactions.

Risk Level	Score Range	Interpretation
Low	0 – 20	Minimal risk signals; majority of all transactions
Medium	21 – 45	Two or more risk signals present; warrants monitoring
High	46 – 70	Multiple strong signals; should be flagged for review

Critical	71 – 100	**All or nearly all signals active; immediate investigation required**

Phase 6 — Summary Statistics & Correlation Analysis

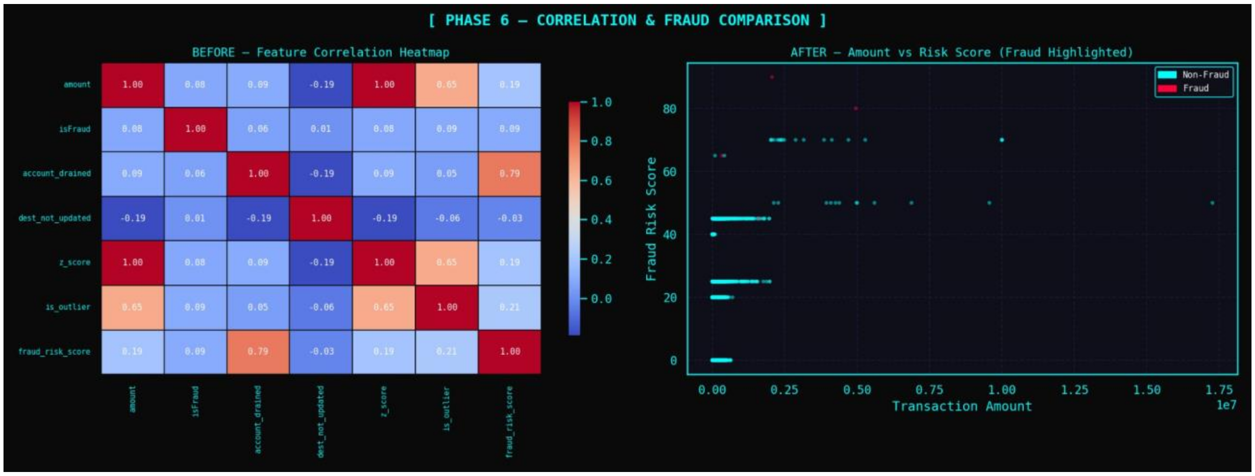
The last stage summarized all the findings and made them available in the form of summary statistics, correlation tables, and graphics in the six analytical dimensions.

Overall Dataset Summary Statistics

Column	Mean	Min	Max
amount	179,861.90	0.00	92,445,516.00
oldbalanceOrg	833,883.10	0.00	59,585,040.37
newbalanceOrig	855,113.67	0.00	49,585,040.37
oldbalanceDest	1,100,701.67	0.00	356,015,889.35
newbalanceDest	1,224,996.40	0.00	356,179,278.92
fraud_risk_score	23.51	0.00	100.00
z_score	0.30	-0.30	152.98

Visualizations Generated

- Fraud vs Legitimate Transaction Distribution (Bar Chart)
- Transaction Amount by Type (Box Plot)
- Fraud Count by Transaction Type (Bar Chart)
- Z-Score Distribution with Outlier Threshold (Histogram)
- Fraud Risk Score Distribution (Dual Panel: Histogram + KDE)
- Correlation Heatmap of All Numeric Features (amount, balances, isFraud, z_score, is_outlier, fraud_risk_score)
- User Spending Behavior (High-Risk vs Normal Users)



4. Conclusions & Recommendations

4.1 Conclusions

The six-phase analysis of 6.36 million PaySim financial transactions yielded the following key conclusions:

- Fraud is extremely rare (0.13%) but concentrated exclusively in CASH_OUT and TRANSFER transaction types. Any detection system should focus filtering efforts on these two categories.
- Balance arithmetic inconsistencies are pervasive over 5 million sender mismatches and nearly 3.9 million receiver mismatches suggest systematic issues in balance updates that correlate strongly with fraudulent activity.
- Z-score outlier detection successfully flagged 44,945 high-value transactions, of which 20.53% are confirmed fraudulent demonstrating the value of statistical anomaly detection as a primary signal.
- The engineered fraud risk score provides a composite 0–100 risk measure that blends five independent signals, enabling ranked prioritization of suspicious transactions for human review.
- With 1,542,928 high-risk users identified from 6.35 million total users (~24.3%), the behavioral user-level grouping provides an effective second layer of filtering beyond per-transaction analysis.

4.2 Recommendations

Recommendation	Rationale
Focus real-time monitoring on CASH_OUT and TRANSFER	100% of fraud occurs in these two types; other types are safe
Flag transactions with fraud_risk_score >= 65	High composite scores combine multiple confirmed risk signals

Investigate balance mismatch anomalies	5M sender mismatches may indicate data pipeline or fraud masking issues
Apply threshold alerts for amounts > 1,991,460	This is the statistical Z>3 boundary where fraud probability rises sharply
Use user-level risk profiles in real-time scoring	Behavioral history improves per-transaction predictions
Consider ML models trained on risk features	The engineered features (risk score, z-score, drain flag) are strong ML inputs

4.3 Technology Stack Used

Library	Version	Purpose
Python	3.13	Primary programming language
Pandas	3.0.1	Data loading, cleaning, grouping, and transformation
NumPy	2.4.3	Mathematical operations, z-score computation
Matplotlib	Latest	Chart and graph generation
Seaborn	0.13.2	Correlation heatmaps and styled visualizations
Jupyter	Latest	Interactive notebook environment in VS Code

End of Report — Fraud Detection Data Analysis System | PaySim Financial Dataset